

OND/TDA Observability Stack

Unified one-page method figure: ODD-OBS (π registry) + OND-ART (structural metrics) + ks1/ks2 orbit-spectrum + topology (TDA)

Figure 1. Pipeline (artifacts $\rightarrow$ canonical observation $\rightarrow$ multi-channel diagnostics $\rightarrow$ report)

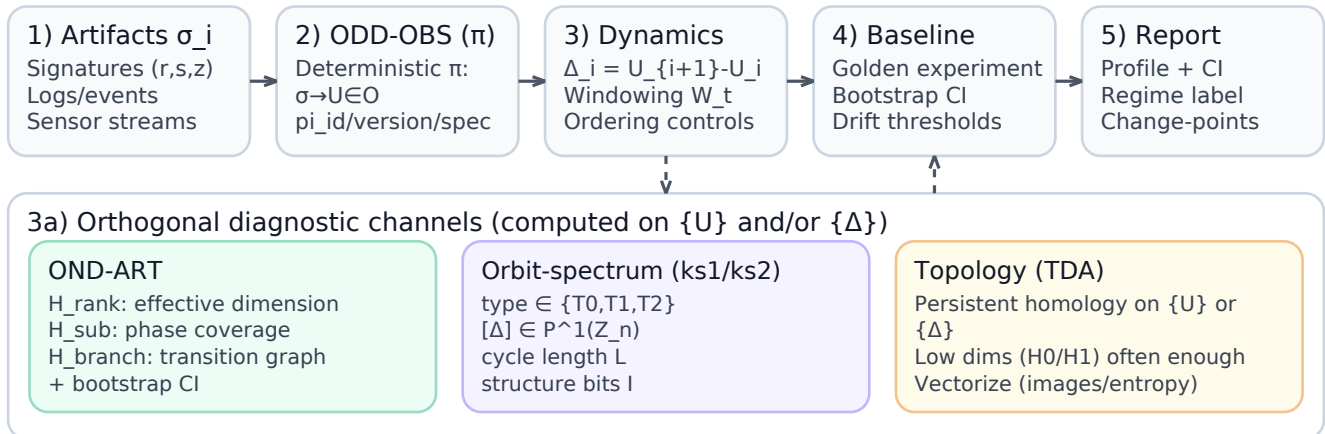


Table 1. Qualitative signatures by channel (validate vs baseline; π fixed)

Phenomenon / defect class	OND-ART (H_{rank} , H_{sub} , H_{branch})	Orbit-spectrum (ks1/ks2)	Topology (TDA) (H_0/H_1 features)	Controls / notes
IID-like / healthy Maximal coverage	High effective dimension; wide phase coverage; no graph bottlenecks	Mostly T0; broad direction spectrum; high $H(\Sigma)$	No stable non-trivial cycles; signatures stable across windows	Compare to baseline CI; keep π unchanged
Low-rank coupling Correlated steps	$H_{rank} \downarrow$ (dimension collapses); often $H_{sub} \downarrow$	Few dominant directions; short cycles	Thin manifold / filament structure; reduced complexity	Check preprocessing; confirm no hidden rescale
Finite-state / throttling Limited transitions	$H_{branch} \downarrow$ (few transitions); possible clustering	Repeated Δ support; compressed spectrum	Graph-like structure; stable components	Inspect caching/fallback logic; correlate with runtime state
Dual-step / bimodality	Two-mode Δ ; branching may $\uparrow$	Type T1; two step clusters; measurable imbalance	Two intertwined filaments; window alternation	Validate via Golden experiment (controlled injection)
State-dependent regime Switching / coupling to U	Window-to-window drift; non-stationary profile	Type T2; Δ depends on state	Topology changes with time; change-points detectable	Test order-invariance; isolate regimes; stratify by context
Entropy starvation / resets Environment-driven	Abrupt drops in H_{sub}/H_{rank} ; repeated patterns	Spikes in short cycles; spectrum collapses	Abrupt simplification; strong change-point signal	Correlate with host metrics; ensure reproducibility

Non-goals (explicit boundaries)

No cryptanalysis, no key/state recovery, no exploitation claims.
 Not a substitute for raw RNG bit tests; complements them at the protocol/trajectory layer.
 No binary pass/fail: outputs are profiles with uncertainty; interpret only vs baseline.